

Student Performance Prediction Using Machine Learning

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1 Introduction

Machine Learning plays an important role in predicting future outcomes based on historical data. In education, predictive models can help identify the factors that influence student academic performance and support data-driven decision-making.

This project focuses on predicting student performance using several machine learning algorithms. The complete workflow includes data preprocessing, exploratory data analysis, model development, evaluation, and interpretation of results.

2 Objective

The objectives of this project are:

- Load and explore the student performance dataset.
- Perform data preprocessing.
- Select relevant features for prediction.
- Conduct exploratory data analysis.
- Train multiple machine learning models.
- Compare model performance.
- Identify the factors that most influence student performance.

3 Dataset Description

The dataset contains approximately one million student records.

The available variables include:

- Student ID
- Weekly Self Study Hours
- Attendance Percentage

- Class Participation
- Total Score
- Grade

The target variable selected for prediction is:

- Total Score

The input features used for training are:

- Weekly Self Study Hours
- Attendance Percentage
- Class Participation

4 Tools and Technologies

The project was developed using:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

5 Methodology

5.1 Data Collection

The dataset was loaded using Pandas and inspected using:

- Dataset dimensions
- Data types
- Missing values
- Duplicate records
- Summary statistics

5.2 Data Preprocessing

Several preprocessing steps were completed before model training.

These included:

- Missing value inspection
- Duplicate checking
- Data type verification
- Feature selection
- Train-test split

No categorical variables were present in the dataset, therefore encoding was not required.

5.3 Exploratory Data Analysis

Several visualizations were created to better understand the relationships between variables.

The analysis included:

- Distribution of Total Score
- Study Hours vs Total Score
- Attendance vs Total Score
- Class Participation vs Total Score
- Correlation Analysis

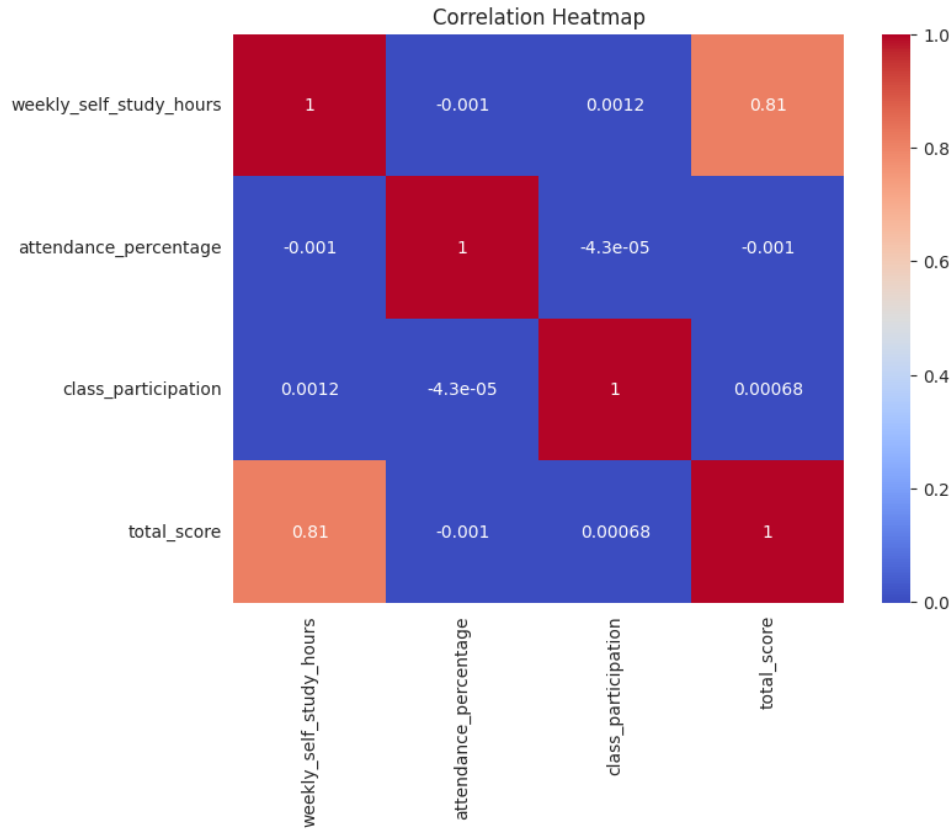


Figure 1: Correlation Heatmap

6 Machine Learning Models

Three regression models were implemented:

1. Linear Regression
2. Decision Tree Regressor
3. Random Forest Regressor

Each model was trained using the same training dataset and evaluated using the same testing dataset.

7 Model Evaluation

Model performance was evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2 Score)

The comparison of model performance is presented below.

Model	MAE	RMSE	R^2 Score
Linear Regression	7.16	9.00	0.6600
Decision Tree	8.37	11.63	0.4319
Random Forest	6.68	9.06	0.6549



Figure 2: Model Comparison by R^2 Score

8 Feature Importance

Feature importance was calculated using the Random Forest model.

This analysis shows which variables contribute the most to predicting student performance.

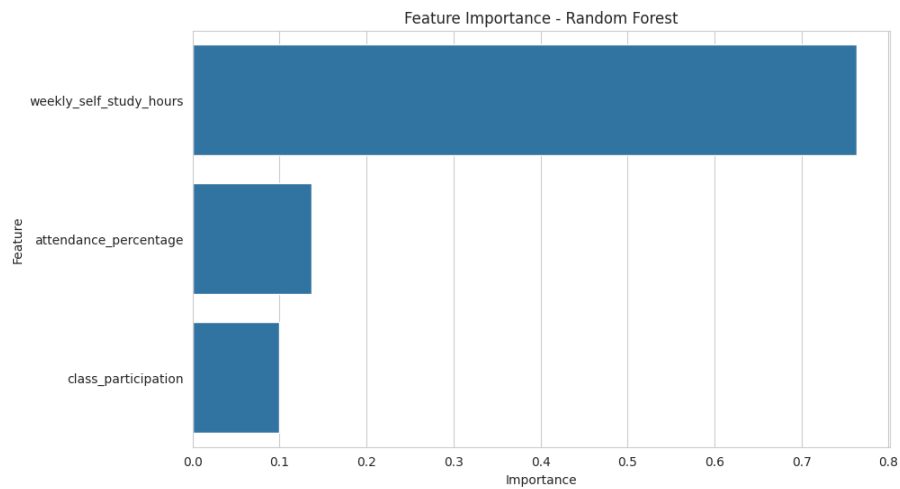


Figure 3: Feature Importance

9 Actual vs Predicted Results

The Actual vs Predicted plot compares the real student scores with the predicted values generated by the Random Forest model.

Points located close to the diagonal indicate better prediction accuracy, while points farther away represent prediction errors.



Figure 4: Actual vs Predicted Total Score

10 Results and Discussion

The experimental results indicate that all three machine learning models were able to predict student performance with reasonable accuracy.

Linear Regression achieved the highest R^2 Score (0.6600), explaining approximately 66% of the variation in students' total scores.

Random Forest produced the lowest Mean Absolute Error (6.68), indicating the smallest average prediction error among the evaluated models.

Decision Tree showed lower predictive performance compared to the other two models.

The analysis also demonstrated that weekly self-study hours, attendance percentage, and class participation are important factors influencing academic performance.

11 Conclusion

This project successfully demonstrated a complete machine learning workflow for predicting student performance.

The project included data preprocessing, exploratory data analysis, feature selection, model development, evaluation, and interpretation of results.

Among the evaluated models, Linear Regression achieved the highest explanatory power, while Random Forest achieved the lowest prediction error.

Overall, the project strengthened practical skills in machine learning, regression analysis, data visualization, and model evaluation. These techniques can be applied to educational analytics and other predictive data science applications.

12 GitHub Repository

The complete source code, notebook, dataset, visualizations, and project files are available on GitHub:

<https://github.com/Nazzekax/student-performance-prediction>